IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-141

1. Topic (12 words max)

Smart Automation of Life Insurance Using AI and Multimodal Interfaces

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Union Assurance, a leading life insurance provider in Sri Lanka, currently operates with a heavily manual process across policy recommendation, customer engagement, sanctions screening, and claims handling. This outdated model results in inefficiencies, errors, and delayed service delivery, directly impacting customer satisfaction and compliance standards.

A critical flaw in the current process is the absence of automated risk assessment. Life insurance underwriting is based on fragmented data and generic scoring models, often ignoring unstructured or handwritten medical documents. This limits personalization and compromises the accuracy of risk evaluations [1][2]. Furthermore, while some frameworks address health risk and pandemic policy design, few propose concrete, AI-based underwriting solutions [2][3].

Policy recommendation relies entirely on human agents, with no automated system to personalize suggestions based on customer profiles. Customers also lack direct, self-service communication channels, depending instead on intermediaries for updates or issue resolution. This leads to a lack of transparency and an inconsistent service experience. Traditional recommendation systems in the industry further exacerbate this issue by functioning as opaque black boxes with no explanation for the options presented [4]. Despite interest in using conversational AI for customer interaction, many existing solutions remain conceptual, offering limited interactivity or personalization [5][6].

Sanction screening is also handled manually, with employees cross-checking national ID numbers against global watchlists. This method is not only inefficient but highly error-prone, making the company vulnerable to compliance violations. While the broader role of insurance in sanctions regimes has been explored [7], current systems rarely use biometric verification or AI-enhanced name-image correlation to prevent false positives and negatives. Advanced methods like fuzzy matching, anti-spoofing detection, and multilingual NLP are largely absent in operational models.

Additionally, the claims process remains email- or agent-driven, resulting in delayed responses, especially during emotionally sensitive periods such as post-bereavement. Current implementations rely on FAQs or static templates and lack intelligent, emotionally aware automation [8][9]. Communication across platforms like SMS, email, and messaging apps is disjointed, increasing response time and reducing user satisfaction.

This research seeks to design an integrated, AI-powered system that automates policy recommendations, risk assessment, sanctions compliance, and claims processing. The solution will leverage technologies like OCR, NLP, facial recognition, and generative AI to reduce turnaround time, improve decision accuracy, and ensure regulatory compliance—ultimately enhancing customer satisfaction and operational resilience.

[1] Evaluation of Risk Level Assessment Strategies in Life Insurance: A Review of the Literature. Available at: https://www.researchgate.net/publication/379093603

[2] Risk Analysis in Healthcare Organizations: Methodological Framework and Critical Variables. Available at: https://www.researchgate.net/publication/327882872

[3] Private Efforts, Public Test Policy and Insurance Against Pandemic Health Risks. ScienceDirect. Available at: https://www.sciencedirect.com/science/article/pii/S0167629621000577

[4] Laurent Lesage, Madalina Deaconu, Antoine Lejay, Jorge Augusto Meira, Geoffrey Nichil, et al.. A Recommendation System For Insurance Built With A Multivariate Hawkes Process Based On Customers' Life Events. 2024. ⟨hal-03483812v2⟩

[5] Lee, Mengwei. "A Life Insurance Policy Bundling Recommendation System Masters Student Paper Competition Submission." *SSRN Electronic Journal*, 1 Jan. 2025, papers.ssrn.com/sol3/papers.cfm?abstract_id=5232331, https://doi.org/10.2139/ssrn.5232331

[6] Z. Khan and A. Roy, "ChatGPT-based Personalized Recommendation System for Health Insurance," *arXiv preprint arXiv:2401.11441*, Jan. 2024. [Online]. Available: https://arxiv.org/abs/2401.11441

[7] Dr George Leloudas, "The role of insurance in sanctions regimes," *ResearchGate*, Oct. 2020. [Online]. Available: https://www.researchgate.net/publication/346173336_The_role_of_insurance_in_sanctions_regimes

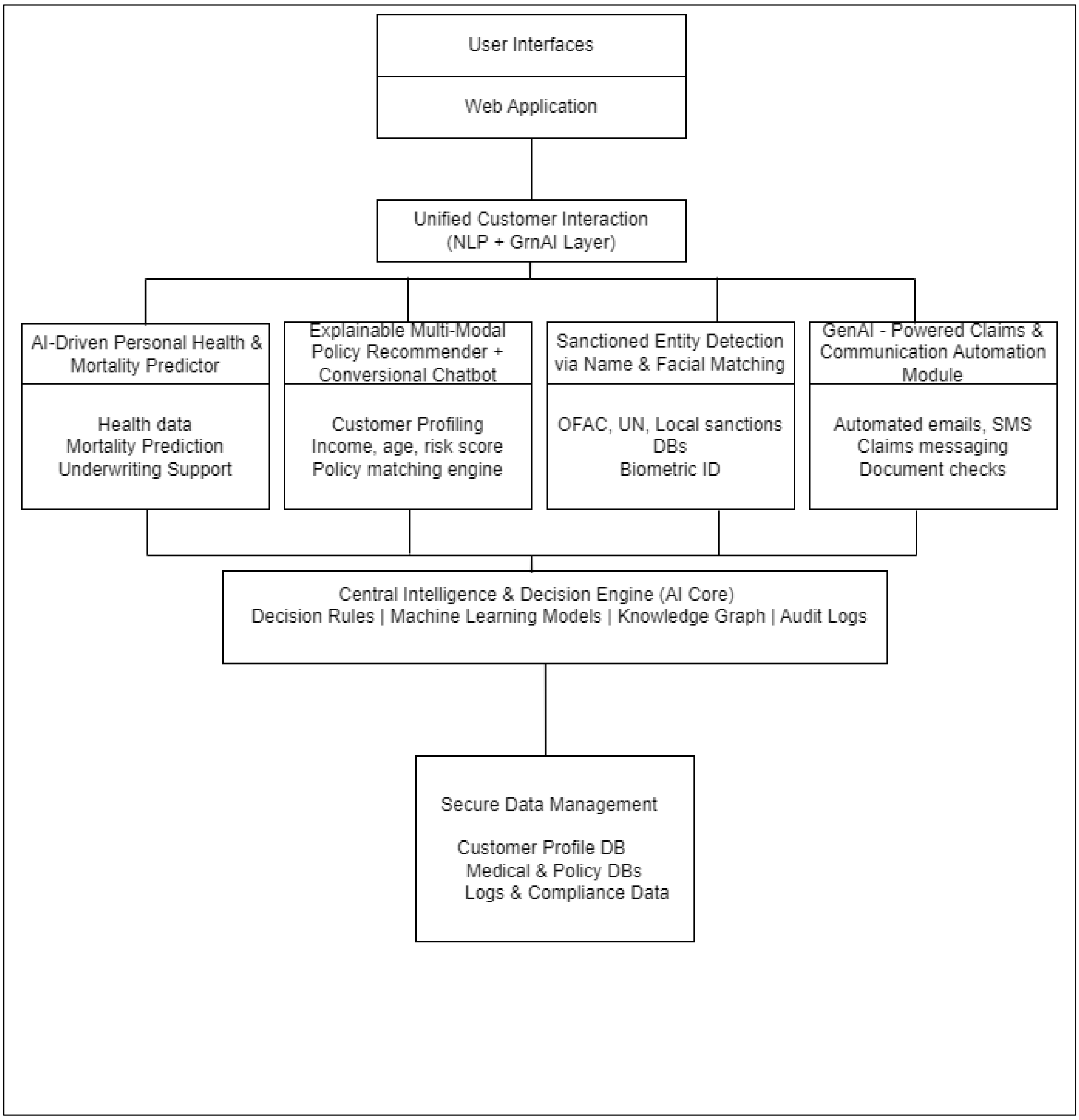
[8] Adavelli Sateesh Reddy, "A Conceptual and Ethical Exploration of Human-Machine Shared Agency," *Philosophical Papers*, 2021. [Online]. Available: https://philpapers.org/rec/SATACP-2

[9] Wiktorsson, Max, "Exploring the Use of Conversational AI in Insurance Services," *Uppsala University Publications*, Jun. 2023. [Online]. Available: https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1899570

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed system aims to transform the manual, error-prone processes in Union Assurance's life insurance operations into a fully automated, intelligent, and customer-centric digital platform. The solution is structured around four core components that work together seamlessly to ensure efficiency, accuracy, and enhanced customer satisfaction.

1. **AI-Driven Personal Health Risk and Mortality Predictor**: This module uses AI models trained on demographic and health data to assess an individual's risk level and predict mortality likelihood. It ensures that underwriting decisions are data-driven, fair, and aligned with the company's long-term risk strategies.
2. **Explainable, Multi-Modal Policy Recommender and Conversational AI Agent for Insurance Personalization**: Unlike current manual agent-based suggestions, this intelligent AI Agent considers multiple customer-specific factors—age, income, risk score, family history, etc. to recommend suitable policies. The use of explainable AI ensures transparency, helping customers understand the reasoning behind recommendations.
3. **Comprehensive Sanctioned Entity Detection Using Multimodal Biometric and Identity Attribute Recognition**: To comply with AML/CTF regulations, this module automatically verifies customer identities against global and local sanction lists using both name-based searches and facial recognition, ensuring secure and compliant onboarding.
4. **Intelligent GenAI-Powered Communication and Claims Automation:** This component revolutionizes customer engagement and claims handling by integrating Generative AI (GenAI), Natural Language Processing (NLP), and intelligent workflow automation. It supports natural, context-aware, two-way communication across channels like Email, SMS, and WhatsApp—ensuring a consistent, empathetic, and responsive user experience. The system automates premium reminders, document validation, inquiry handling, and claims communication, significantly reducing processing time and operational overhead while enhancing transparency and trust throughout the policy lifecycle.


User Interfaces
Web Application
Unified Customer Interaction
(NLP + GrnAI Layer)
AI-Driven Personal Health &
Mortality Predictor
Health data
Mortality Prediction
Underwriting Support
Explainable Multi-Modal
Policy Recommender +
Conversional Chatbot
Customer Profiling
Income, age, risk score
Policy matching engine
Sanctioned Entity Detection
via Name & Facial Matching
OFAC, UN, Local sanctions
DBs
Biometric ID
GenAI - Powered Claims &
Communication Automation
Module
Automated emails, SMS
Claims messaging
Document checks
Central Intelligence & Decision Engine (AI Core)
Decision Rules | Machine Learning Models | Knowledge Graph | Audit Logs
Secure Data Management
Customer Profile DB
Medical & Policy DBs
Logs & Compliance Data

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The proposed solution requires interdisciplinary expertise across insurance, artificial intelligence, natural language processing (NLP), and regulatory compliance domains.

1. **Insurance Domain Expertise**: Understanding life insurance underwriting principles, customer risk profiling, policy structuring, and claims management is essential. This ensures that automated recommendations and claim decisions align with real-world business rules and legal frameworks.
2. **AI and Machine Learning Expertise**: Developing accurate health risk and mortality prediction models involves applying supervised learning techniques, using relevant medical, demographic, and behavioral datasets. Domain-specific feature engineering, model tuning, and evaluation metrics are critical for delivering actionable predictions.
3. **Natural Language Processing (NLP) and Explainable AI**: Building an explainable multi-factor AI Agent requires expertise in NLP models capable of handling multilingual, context-aware communication. Explainable AI (XAI) methods must be integrated to ensure transparency in policy recommendation reasoning.
4. **Computer Vision and Sanction Screening**: The sanctioned entity detection module involves facial recognition and name matching against global and national watchlists. This requires deep learning expertise in computer vision, biometric validation, and entity resolution.
5. **Generative AI and Claims Automation**: The intelligent two-way communication system leverages GenAI for dynamic, human-like interactions. It uses OCR and NLP for document analysis (e.g., prescriptions, bills), automated claim validation, sentiment scoring, and escalation handling based on intent and emotional tone.
6. **Regulatory and Ethical Compliance**: The system must align with local and international regulations (e.g., AML/CTF laws, data privacy acts). Experts in legal compliance and ethics in AI must guide the solution design.
7. **Data Requirements**: Key data sources include historical customer and claims data, anonymized health records, global sanction databases (e.g., OFAC, UN), facial datasets for verification, and multilingual text corpora for AI Agent training. Data privacy, security, and consent handling are fundamental during data collection and processing.

Together, these competencies ensure a reliable, secure, and customer-centric automation framework for life insurance operations.

8. Objectives and Novelty

Main Objective The main objective of this research is to design and develop an intelligent, fully automated life insurance management system that enhances accuracy, efficiency, and customer satisfaction. By integrating AI-driven risk prediction, explainable policy recommendation and AI Agent, automated sanction screening, and GenAI-powered two-way communication and claims processing, the system aims to replace outdated manual workflows. The goal is to ensure data-driven decision-making, faster and more transparent claim settlements, and proactive, personalized customer engagement ultimately modernizing the life insurance process and aligning it with regulatory and service excellence standards.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Kasthuriarachchi N.K.A.T.L IT22554772	Intelligent Two-Way Communication & Claims Automation System for Life Insurance	**GenAI Message Generation** • Implement **Generative AI models** for crafting dynamic, personalized, and context-aware messages. • Configure **multi-stage tone control** (friendly > warning > final notice).	**Life Insurance-Specific Focus** • Unlike most systems optimized for general or P&C (Property and Casualty) insurance, this solution is tailored to the emotional, procedural, and regulatory nuances of life insurance, including sensitive communications and longer relationship cycles.

		Notification and Escalation Logic • Develop time-based triggers for automated notifications (before due, on due, overdue stages). • Include escalation rules for high-risk cases, delayed payments, and emotional tone in replies. **Customer Behavior Analysis** • Implement a profiling engine to assess loyalty, claim history, and payment behavior. • Use this to adjust frequency, tone, and urgency of communications. **Claims Document Processing and Validation** • Use OCR + NLP to extract, compare, and validate data from prescriptions, bills, and claim documents. • Apply coverage rules and fraud detection algorithms for automatic claim triaging.	**Two-Way Conversational GenAI** • Introduces a truly interactive GenAI mail system that not only pushes notifications but understands and replies in natural language, handling diverse queries like “Cancel my policy”, “I’ve paid already”, and “Where’s my claim?” with empathy and intelligence. **Behavior-Based Personalization** • Goes beyond rule-based automation by using payment history, loyalty, and past interactions to adapt message tone, frequency, and urgency, creating a more human and respectful customer experience.

		Automated Claims Decision Engine • Automate straightforward claim approvals/rejections with GenAI-generated justifications. • Flag edge cases for human review with summarized input.	**Intelligent Claims Validation with OCR + NLP** • Enhances document-based claims processing with automated parsing and matching of prescriptions and bills—something few life insurance platforms currently do, especially using AI to validate, detect fraud, and auto-approve simple cases. **Emotion-Sensitive Escalation and Support** • Implements sentiment analysis and emotional tone detection in user replies, allowing early escalation of distressed or dissatisfied customers to human agents—addressing a key limitation in current customer service bots.

			Seamless Communication-to-Claims Bridge • Uniquely unifies premium customer support, and claims automation into one intelligent loop—ensuring continuous engagement and context-aware support throughout the policy lifecycle.
Rashmika D.G.C.D IT22369338	Comprehensive Sanctioned Entity Detection Using Multimodal Biometric and Identity Attribute Recognition	**Data Source Integration** • Aggregate and maintain access to international watchlists (OFAC, UN, EU, etc.). • Automate periodic updates of watchlist databases to ensure current data availability. **Textual Name Screening Module** • Implement fuzzy name matching algorithms for variant detection (typos, aliases). • Integrate NLP-based multilingual name recognition to support global applicability.	**Biometric Verification Layer** • Unlike most existing systems that rely only on name and textual data, this research integrates facial recognition technology for enhanced identity verification—an innovative and underutilized approach in insurance sanctions screening.

		Biometric Facial Recognition Module • Integrate facial recognition technology to compare customer photos with watchlist images. • Develop preprocessing pipelines to handle image quality, format normalization, and resolution issues. **Multimodal Identity Matching Engine** • Combine name and facial recognition outputs using a unified risk score. • Design logic for prioritization and flagging of high-risk entities. **Anti-Spoofing and Image Verification** • Implement liveness detection, deepfake detection, and forgery detection mechanisms. • Use computer vision models trained to spot manipulate synthetic identities.	**Multimodal (Name + Face) Matching** • Proposes a combined analysis of textual and biometric data, significantly reducing both false positives and false negatives—a major improvement over uni-modal matching systems. **Advanced Anti-Evasion Mechanisms** • Introduces deepfake detection, forged image analysis, and anti-spoofing tools—novel enhancements not found in standard compliance systems, addressing modern threats.

		Real-Time Processing Framework • Design system to support real-time screening during onboarding, claim filing, or policy issuance. • Optimize for low-latency response in multilingual, international environments. **Insurance-Specific Workflow Integration** • Tailor system modules to integrate with common insurance workflows (e.g., underwriting, claims). • Ensure audit trails, documentation, and alerts align with insurance compliance standards.	**Real-Time, Multilingual, Multimodal Screening Architecture** • Develops a real-time system capable of processing both names and images across multiple languages and character sets—currently missing in conventional insurance solutions. **Tailored for Insurance Domain** • Focuses uniquely on insurance-specific scenarios like policy onboarding and claims processing, while most existing research focuses only on banking or payments. This fills a critical research and operational gap.

Methyani K.A.G.C IT22330246	AI-Driven Personal Health Risk and Mortality Predictor for Life Insurance Underwriting	**OCR Pipeline Development** • Build and fine-tune OCR models for digitizing handwritten and scanned medical records. • Validate accuracy and reduce noise/errors in text extraction, especially for handwritten doctor notes and prescriptions. **Medical NLP Engine** • Design NLP algorithms to extract medical entities (conditions, symptoms, treatments). • Classify extracted conditions with severity tags (e.g., stage 1 diabetes vs. stage 4 cancer). • Create rule-based and ML-based severity scoring dictionaries using medical taxonomies (e.g., ICD-10, SNOMED CT).	**Advanced Handwritten Medical Record Digitization** • Most existing systems exclude or poorly handle handwritten data, resulting in partial applicant profiles. • This solution incorporates a dedicated OCR module for handwritten and scanned records, allowing full digitization of doctor notes, discharge summaries, and legacy medical documents assuring no critical data is lost in the underwriting process.

		Data Integration & Standardization • Merge clinical, diagnostic, and lifestyle data into a unified structured dataset. • Clean, normalize, and standardize inputs for ML compatibility (e.g., missing value imputation, unit conversions). **Mortality Risk Modeling** • Train predictive models (e.g., Gradient Boosting, LSTM, XGBoost) to estimate: ❖ Mortality risk ❖ Life expectancy range ❖ Risk evolution over time • Calibrate models using labeled datasets with real underwriting or actuarial outcomes.	**Severity-Based Health Risk Analysis (Beyond Binary Classification)** • Traditional models often provide binary classifications (e.g., "at risk" or "not at risk"). • This research introduces nuanced, stage-based severity tagging (e.g., "Stage 4 lung cancer", "mild hypertension") to deliver granular risk profiles. • Enables more accurate and fair premium pricing by accounting for condition intensity, not just presence.

		Underwriter Dashboard Development • Design a clean, interactive dashboard to display: ❖ Extracted medical conditions and their severity ❖ Personalized life expectancy prediction ❖ Risk confidence scores and visualizations **Impact Measurement Module** • Develop metrics for: Processing speed before/after automation Fairness indicators (e.g., demographic parity) Prediction accuracy against benchmarked actuary decisions	**Holistic, Multi-Source Risk Profiling** • Unlike fragmented systems that use either clinical or lifestyle data, this solution integrates multiple data dimensions—medical records, physical activity, nutrition, smoking habits, alcohol use into a comprehensive applicant health profile. • This holistic view enhances predictive performance and underwriting fairness. **Personalized Mortality Estimation with Transparency** • Provides individualized life expectancy ranges rather than broad actuarial group averages. • Combines explainable ML models with visual dashboards, allowing underwriters to understand the 'why' behind each prediction—an element often missing in black-box systems.

		User Testing and Continuous Feedback • Conduct A/B testing with underwriters and actuarial teams. • Collect feedback to iteratively improve model transparency and dashboard usability.	**Unified Underwriting Interface** • Replace fragmented toolchains with a single interactive dashboard showing medical NLP output, severity scoring, predicted mortality, and action triggers—drastically improving decision speed, consistency, and auditability. **Quantifiable Impact Reporting** • Embeds an impact measurement module to quantify improvements in: ❖ Underwriting time ❖ Decision consistency ❖ Pricing fairness ❖ Model explainability
Mallawaarachchi P.K. IT22088178	Explainable, Multi-Modal Policy Recommender and Conversational AI Agent for Insurance Personalization	**Data Collection & Preprocessing** • Parse financial records to determine income, expenses, and affordability. • Gather demographic data through structured form-based inputs (age, location, occupation, etc.).	**Explainable White-Box Recommendation** • Unlike black-box models, this system articulates human-readable reasons for every policy recommendation, increasing user trust and decision clarity.

		Feature Extraction & Profiling • Conduct **affordability assessment** based on income vs. policy premium structure. • Apply **clustering and segmentation** techniques on demographic attributes to group similar customers. **Policy Clustering & Recommendation Engine** • Cluster available policies based on target user groups (e.g., health, affordability, family structure). • Match customer profiles to suitable clusters and compute **multi-factor scores** for top policy recommendations. • Rank and filter policies using scores for medical, financial, and demographic compatibility.	**Multi-Modal Data Fusion** • Integrates medical data, financial profiles, and demographics—a unique, holistic customer profiling approach rarely seen in traditional systems. **Intelligent and Interactive AI Agent Interface** • Includes a standalone, intelligent AI Agent that interacts with users through natural conversation, allowing them to ask questions like *“Why not Policy B?”* or explore ways to reduce premiums—offering a novel conversational layer beyond the core recommendation engine.

		Explainability Module • Generate **natural-language justifications** explaining why a particular policy is recommended. • Implement rule-based and interpretable machine learning models to support explainable outputs. • Handle “why not” or “what if” scenarios for comparative policy explanations. **Interactive AI Agent Interface** • Develop a conversational AI Agent that supports **Q&A**, user challenges, and follow-up queries. • Display **reason-based policy alternatives** using concise, human-understandable summaries. • Log user feedback to refine recommendation logic and explanation clarity.	**Policy Clustering for Transparent Matching** • Clusters policies based on similarity in coverage, price, and eligibility, then match users with clear dimension-wise scoring—offering a layered, intelligible match process. **Application to High-Stakes Personalization** • Tailored specifically for sensitive use in insurance, with potential scalability to healthcare, finance, or legal services, where explainability is essential for compliance and ethics.

		Evaluation and Validation • Conduct experiments comparing **black-box** vs. **white-box** models in terms of trust and satisfaction. • Measure **user comprehension**, perceived transparency, and **recommendation accuracy**. • Assess scalability and applicability to other sensitive domains (e.g., legal or healthcare).	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Kasthuriarachchi N.K.A.T.L IT22554772	This component introduces a transformative approach to how life insurance providers manage customer interactions and claims. By combining Generative AI (GenAI), Natural Language Processing (NLP), and intelligent automation, it enables natural, two-way communication while streamlining internal workflows. The solution ensures timely premium tracking, personalized and empathetic customer engagement, automated document validation, and responsive claim handling. It operates seamlessly across multiple communication channels—Email, SMS, and WhatsApp—offering a consistent and unified user experience from policy initiation to claim settlement.
Rashmika D.G.C.D IT22369338	This component introduces a next-generation customer screening system that enhances sanctions compliance for life insurance providers by integrating biometric verification with AI-driven name matching. Unlike traditional systems that rely solely on textual data (e.g., names, dates of birth, and nationalities), this solution incorporates facial recognition technology to detect high-risk individuals even when aliases, misspellings, or evasion tactics are used. The system performs real-time, multimodal screening by comparing customer names and facial images against international watchlists such as OFAC, UN, and EU sanctions databases. Advanced features include fuzzy name matching, multilingual NLP for name recognition across languages, and risk scoring based on name-image correlation. To further mitigate risk, the system employs anti-spoofing techniques to detect manipulated images, such as deep fakes or forged IDs. Tailored specifically for the insurance domain, this component supports unique workflows such as policy issuance, renewal, and claims, ensuring compliance is upheld throughout the customer lifecycle. By drastically reducing false positives and catching potential false negatives, it improves operational efficiency and protects insurers from reputational and regulatory harm. The integration of biometric and textual screening, combined with real-time global processing, marks a significant advancement in intelligent, secure, and compliant insurance practices.
Methyani K.A.G.C IT22330246	This component introduces a next-generation AI underwriting system that aims to revolutionize life insurance risk evaluation by delivering personalized, data-driven assessments. Traditional underwriting often struggles with fragmented data sources, especially handwritten or unstructured medical documents, leading to generalized risk scoring. This

	system overcomes those barriers by integrating Optical Character Recognition (OCR), Natural Language Processing (NLP), and predictive analytics into a cohesive pipeline. Handwritten and scanned medical records are first digitized using advanced OCR techniques. NLP models then extract relevant health indicators, diagnoses, and patterns from both structured and unstructured data. These insights are fed into a machine learning model trained on mortality and morbidity risk factors, enabling the system to produce accurate, individualized risk profiles. The result is a fully automated underwriting process that is faster, more accurate, and tailored to each applicant's unique medical and demographic context. This not only enhances precision underwriting but also promotes fairness and transparency in premium determination. By minimizing manual intervention and standardizing risk assessment, this component significantly improves operational efficiency and customer experience for insurers.
Mallawaarachchi P.K. IT22088178	This component integrates a generative AI-powered AI Agent with an intelligent recommendation engine to offer transparent, personalized life insurance policy suggestions. By leveraging multi-modal data such as the user's medical history, financial capacity, and demographic details the system generates tailored policy options aligned with the user's risk profile and preferences. What sets this component apart is its emphasis on explainability and interaction. Unlike traditional recommendation systems that act as black boxes, this AI Agent communicates the reasoning behind each suggestion in natural language. Users can ask follow-up questions, request comparisons between policies, and gain clarity on premium costs, benefits, and exclusions. This fosters user trust and informed decision-making, particularly important in high-stakes domains like health and financial insurance. The system architecture combines rule-based logic, machine learning classifiers, and NLP techniques to ensure accuracy, fairness, and transparency. By integrating this component into the broader insurance process, companies can significantly enhance customer satisfaction and reduce dropout rates during policy selection.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Assistant Professor	Dr. Lakmini	Abeywardhana	
Co-Supervisor	Assistant Lecturer	Ms. Malithi	Nawarathne	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

I recommend this project to carry out as final year Datascience research topic.

IT4010 Research Project – 2025 July **Topic Assessment Form**

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.